🛑 Important : Terms & Conditions

Terms & Conditions – Confidentiality & Intellectual Property

These Terms & Conditions ("**Terms**") govern participation in the competition ("**Competition**") organized by **Sunrise Mentors Private Limited** ("**Organizer**") in collaboration with **Eternal Limited** ("**Zomato**").

By registering for or participating in the Competition, you ("**Participant**" or "**you**") confirm that you have read, understood, and agreed to be bound by these Terms.

1. Purpose

During the course of the Competition or after the Competition, Participants may receive access to certain proprietary, or sensitive information belonging to Zomato. These Terms define how such information must be handled and clarify ownership of Intellectual Property created or disclosed during the Competition.

2. Eligibility

The Competition is open to students who meet the eligibility criteria specified by the Organizer. Participation is voluntary and does not create any employment, agency, or contractual relationship with Zomato.

3. Confidential Information

3.1 Definition

"**Confidential Information**" includes all the information disclosed directly or indirectly by Zomato or its representatives, in any form, including but not limited to:

- Information shared during interviews, briefings, mentoring sessions, or Competition activities;
- Technical information, software, source code, object code, algorithms, databases, or system designs;
- Business processes, internal communications, methods, or operational data;
- Marketing strategies, financial information, projections, or business plans;
- Any information that a reasonable person would consider confidential;

Confidential Information need not be novel, unique, or patentable.

4. Participant Obligations

By participating in the Competition, you agree to:

- **Confidentiality:** Keep all information strictly confidential and not disclose it to any third party without prior written consent from Zomato
- **Limited Use:** Use the information solely for the purpose of participating in the Competition
- **Protection:** Take all the precautions to safeguard information, at least equal to the care used for your own sensitive information
- **No Unauthorized Copying:** Not record, copy, or reproduce the information except where strictly necessary for Competition participation
- **Breach Notification:** Immediately notify the Organizer and Zomato if you become aware of any unauthorized disclosure or misuse

5. Intellectual Property & Ownership

- **Ownership:** All the information remains the exclusive property of Zomato

- **No License Granted:** Participation does not grant any license or right to use Zomato's Intellectual Property, including patents, copyrights, trademarks, or trade secrets
- **Brand Usage:** Participants may not use Zomato's name, logo, or trademarks without prior written authorization in writing from Zomato
- **Participant Creations:** Any ideas, concepts, inventions, designs, or materials created by a Participant during the Competition that relate to Zomato or are based on information shall be treated as Confidential Information

6. Return or Destruction of Information

Upon request by the Organizer or Zomato, or upon completion of the Competition, Participants must:

- Return or permanently destroy all materials containing the information
- Delete all the information from personal devices and storage systems
- Provide written confirmation of compliance if requested

2. Legal Remedies & Indemnity

- **Injunctive Relief:** Participants acknowledge that unauthorized disclosure may cause irreparable harm, and Zomato may seek injunctive or equitable relief without limitation
- **Indemnity:** Participants agree to indemnify and hold harmless Zomato from any losses, costs, or legal expenses arising from breach of these Terms
- **No Warranty:** All information is provided "as is," without warranty of accuracy or completeness

3. **Term & Survival**

If the Participant does not enter into a separate agreement with Zomato, confidentiality obligations under these Terms shall remain in effect for **one (1) months** from the date the Participant first participates in the Competition or such information is in public domain, whichever is later.

4. Relationship of Parties

Nothing in these Terms creates an employment, partnership, joint venture, or agency relationship between the Participant and Zomato or the Organizer.

5. Acceptance

By signing below or registering for the Competition, the Participant confirms acceptance of these Terms & Conditions.

Problem Statement 1

Improving Kitchen Prep Time (KPT) Prediction to Optimize Rider Assignment and Customer ETA at Zomato

Context

Zomato operates a large-scale food delivery ecosystem where delivery time accuracy is critical to customer experience, rider efficiency, and platform costs. A key component of delivery ETA is **Kitchen Prep Time (KPT)** — the time taken by a restaurant to prepare an order after it is confirmed.

Accurate KPT prediction enables better rider assignment, reduces idle waiting at restaurants, and helps set reliable expectations for customers. Given Zomato's scale (300K+ merchants), even small improvements in KPT accuracy can have significant operational impact.

Business Challenge

Currently, KPT prediction models are primarily trained using **Food Ready** signals marked by merchants on the Zomato Merchant Experience (Mx) app. However, these signals are not always a true reflection of when food is actually ready.

Key limitations of current input signals include:

- **Rider-influenced FOR (Food Order Ready) marking**: Merchants sometimes mark food as ready when the rider arrives, not when preparation is completed
- **No visibility into kitchen-wide rush**: The system only observes Zomato orders, while in-store dining and competitor orders also contribute to kitchen load
- **Human and operational bias**: Manual marking behavior varies significantly across merchants and time periods

These limitations introduce noise and bias into KPT labels, leading to inaccurate predictions, early rider arrivals, increased waiting time, and fluctuating ETAs for customers.

Problem Statement

How can Zomato improve the accuracy of **Kitchen Prep Time (KPT) prediction** by strengthening and enriching input signals beyond merchant-marked FOR, while ensuring solutions are scalable across hundreds of thousands of merchants?

Key Deliverables

Important guidance: Participants should assume that **existing KPT prediction models are largely fixed** and that **model changes alone are not sufficient**. While model improvements may be proposed if strongly justified, the primary focus should be on

improving input signals, data quality, system design, merchant workflows, or hardware-assisted instrumentation that can significantly enhance KPT prediction.

Participants are expected to propose solutions that:

- Improve the accuracy and reliability of input signals feeding the KPT model
- Reduce bias introduced by merchant-marked Food Ready (FOR) events
- Capture kitchen rush and load beyond Zomato-only orders
- Are feasible and scalable across Zomato's merchant base

Purely model-centric solutions without signal or system-level improvements will be considered insufficient.

Success Metrics

- Average rider wait time at pickup
- ETA prediction error (P50 / P90)
- Order delay and cancellation rates
- Rider idle time

Problem Statement 1 - Evaluation Criteria

Evaluation Criteria - Problem Statement 1

For Problem Statement 1, the evaluation would be done on the basis of:

- The problem understanding and the analysis of downstream impact of inaccurate predictions.
- **Signal Quality improvement:** How does the proposal de-noise the current signals or introduce signals into the system.
- How the Restaurant Live Rush (apart from the Zomato orders) can be handled?
- How much does the proposal stress upon the scalability (adhering to both small and large scale restaurants) ?
- **Simulation (bonus points) -** Simulate or build a quantitative/IOT based model that demonstrates significant improvements. Can also propose AI first solutions.
- Impact on the success metric
- **Novelty and Creativity -** How unique is the idea compared to the other solutions.

Penalties:

- If the proposal is entirely model specific, even though we specified that the proposal should contain improvements in improving the FOR signals.
- No connection to success metric: the proposal doesn't map to the success metrics specified in the Problem Statement.

Problem Statement 2

Cart Super Add-On (CSAO) Rail Recommendation System - Problem Statement

Background

In the food delivery business, cart value optimization is critical for both customer satisfaction and business profitability. Currently, when customers add items to their cart, they may miss discovering complementary items that would enhance their meal experience. A significant opportunity exists to intelligently recommend additional items (add-ons) that customers are likely to accept, thereby increasing order value while improving customer experience.

Problem Statement

How can we build an intelligent recommendation system that suggests relevant add-on items to customers based on their current cart composition, contextual factors, and historical behavior patterns, while maintaining high acceptance rates and customer satisfaction?

Current Challenges

1. Cart Context Understanding

- Difficulty in understanding meal composition and identifying complementary items
- Limited ability to recognize incomplete meal patterns (e.g., main dish without beverage, single item orders)
- Lack of personalization based on individual user preferences and past ordering behavior

2. Contextual Relevance

- Recommendations often ignore critical contextual factors:
    - Time of day (breakfast, lunch, dinner, late-night)
    - Restaurant characteristics (cuisine type, price range, specialty items)
    - Geographic location and delivery zone preferences
    - User segments (budget-conscious, premium, frequent orderers)
    - Offer and price sensitivity

3. Business Impact

- Suboptimal cart values leading to missed revenue opportunities
- Low acceptance rates on current recommendation approaches
- Poor targeting leading to recommendation fatigue and negative user experience
- Non-relevant recommendations can impact customer trust.

4. Modelling Challenges

- Recommendation must be real-time and relative to context i.e. with addition of new items,recommendations list must keep on updating to complete the current meal (e.g. Biryani in cart -> recommend salan -> added salan -> now can recommend gulab jamun -> added gulab jamun -> now recommend drinks).

- Latency must be within 200-300 ms to cater to high throughput requests at large scale.

Objectives:

Primary Objective

Develop a scalable recommendation system that accurately predicts which add-on items a customer is most likely to add to their cart, given their current cart state and contextual information.

Success Criteria

The solution should optimize for:

1. Prediction Accuracy: Ability to correctly identify items that users will add vs. reject
2. Business Impact: Measurable increase in below metrics
    a. average order value (AOV)
    b. CSAO rail order share
    c. Cart-to-order (C2O) ratio
    d. CSAO rail attach rate
    e. average number of items ordered
3. User Experience: High acceptance rates of CSAO rail indicating relevant recommendations
4. Scalability: Ability to serve real-time predictions for millions of daily orders within minimal latencies

Key Metrics

Model Performance Metrics - Can include following but not limited to

Metric	Description
AUC (Area Under ROC Curve)	Overall model discrimination ability
Precision @ K	Accuracy of top-K recommendations
Recall @ K	Coverage of relevant items in top-K
NDCG	Ranking quality metric

Business Impact Metrics - Can include following but not limited to

Metric	Description
Add-on Acceptance Rate	% of recommended items added to cart
AOV Lift	Incremental increase in cart value
Click-Through Rate	% of users engaging with recommendations
C2O (cart-to-order) Rate Impact	Effect on order completion

Operational Metrics

Metric	Description
Inference Latency	Time to generate recommendations
Coverage	% of scenarios with predictions
Feature Preparation pipeline	How easy it is to prepare a feature pipeline for training and inference.And how often the feature be refreshed

Input Data Available

You can think of using the features in terms of following entities. But you are free to curate your own features that can create a positive impact of the recommendations.

User Features

- Historical ordering behavior (frequency, recency, monetary value)
- Preferred delivery zones, restaurants, and cuisine types
- User segment information (budget, premium, occasional)
- Session-level engagement metrics

Restaurant Features

- Restaurant characteristics (cuisine, price range, ratings)
- Historical performance metrics (delivery ratings, order volumes)
- Menu composition and item popularity
- Chain vs. independent restaurant indicators

Cart Context

- Current items in cart (dishes, prices, quantities)

- Item-level features (veg/non-veg, dish categories, prices)
- Cart-level aggregates (total value, item count)

Contextual Features

- Temporal context (hour, day of week, meal time)
- Geographic context (city, delivery zone)
- Candidate items to rank (potential add-ons with features)

Historical Interaction Data

- User-item interaction history
- User-restaurant interaction patterns
- Session-level behavioral signals

Expected Output

For each prediction request, the system should return:

- Ranked list of candidate items with probability scores
- Top N recommendations (typically N=8-10) for display
- Response time under latency constraints

Evaluation Framework

Offline Evaluation

- Train-test split on historical data (temporal split recommended)
- Evaluate on multiple metrics: AUC, Precision@K, Recall@K, NDCG
- Perform error analysis on different user segments and contexts

Online A/B Testing (Post-Development)

- Compare against baseline recommendation system
- Monitor business metrics: AOV lift, acceptance rate, CTR
- Track user experience metrics: cart abandonment, order completion
- Analyze performance across different segments and contexts

Constraints & Considerations

Technical Constraints

- **Latency:** Predictions must be served in real-time (< 2-300ms)
- **Scale:** System must handle millions of prediction requests daily
- **Data freshness:** Model should incorporate recent behavioral patterns

Business Constraints

- **Cold start problem:** Handle new users, restaurants, and items
- **Diversity:** Avoid recommending only high-margin items
- **Fairness:** Ensure recommendations work across all restaurant types

User Experience Constraints

- **Relevance:** Avoid irrelevant recommendations that cause annoyance
- **Transparency:** Recommendations should feel natural and contextual
- **Non-intrusiveness:** Don't disrupt the ordering flow

Success Looks Like

A successful solution will:

1. Demonstrate strong predictive performance across diverse scenarios (different meal times,restaurant types, user segments)
2. Show clear business impact through AOV lift and high acceptance rates without degrading user experience
3. Be production-ready with acceptable latency, scalability, and maintainability characteristics
4. Provide actionable insights about what makes certain recommendations successful
5. Generalize well to new restaurants, items, and evolving user preferences

Submission Requirements

Teams should provide:

1. **Model Development**
    a. Data preprocessing and feature engineering pipeline
    b. Model training code with clear documentation
    c. Hyperparameter tuning approach

2. **Evaluation Results**
    a. Performance metrics on holdout test set
    b. Error analysis and insights
    c. Comparison with baseline approaches

3. **Technical Documentation**
    a. System architecture and design decisions
    b. Scalability considerations
    c. Trade-offs and limitations

4. **Business Impact Analysis**
    a. Projected AOV lift and acceptance rates
    b. Segment-level performance breakdown
    c. Recommendations for deployment strategy

Problem Statement 2 - Evaluation Criteria

Evaluation Criteria - Problem Statement 2

Your submissions will be evaluated across six core dimensions. We are looking for end-to-end thinking, from how you handle raw data to how your system drives real business value.

1. **Data Preparation & Feature Engineering (20%)**
    - **Dummy Data Realism:** How well does your synthetic data mimic the messy, real-world reality of food delivery (e.g., city wise behaviour, peak hours, sparse user histories,varying cart sizes, mealtime behaviour etc)?
    - **Contextual Capture:** How effectively do your features capture the dynamic context of a cart (e.g., the transition as items are added or removed) alongside temporal and geographic factors?
    - **Feature Pipeline:** Clarity and feasibility of how features are generated, stored, and retrieved for real-time inference.
    - **Cold Start Strategy:** Distinct data pipelines or fallback heuristics for new users or newly listed restaurants.

2. **Ideation & Problem Formulation (15%)**
    - **Problem Framing:** How have you mathematically and logically framed the recommendation task? (e.g., Are you treating this as a simple classification, a ranking problem, or a sequential task?)
    - **Handling Constraints:** Your conceptual approach to solving the specific challenges outlined in the problem statement (cold start, incomplete meal patterns - may be for some customers we have data for evening and dinner mealtime not for lunch , diversity).

3. **Model Architecture & The "AI Edge" (20%)**
    - **Architecture Rationale:** The justification for your chosen model(s). Why is this approach better suited for this specific Zomato use case than alternative methods?
    - **Handling Sequential Logic:** How well the model accounts for the order and combination of items currently in the cart.
    - **The AI Edge (Bonus/Differentiator):** Extra weight will be given to teams that creatively leverage modern, state-of-the-art Artificial Intelligence (LLMs) to solve complex nuances (e.g., deep personalization, handling unstructured data, or augmenting data sparsity) beyond traditional baseline algorithms.

4. **Model Evaluation & Fine-Tuning Strategy (15%)**
    - **Validation Methodology:** How you split your data for training and testing to prevent data leakage and accurately simulate real-world deployment.
    - **Metric Alignment:** The relevance of your chosen offline evaluation metrics to the core objective of the CSAO rail.
    - **Optimization:** Your strategy for fine-tuning the model, handling trade-offs (e.g., accuracy vs. latency), and your approach to error analysis on underperforming segments.

5. **System Design & Production Readiness (15%)**
    - **Latency Budget:** A clear, architecturally sound explanation of how your system will return predictions within the strict 200-300ms constraint.
    - **Scalability:** How your proposed architecture scales to handle millions of daily prediction requests during peak lunch/dinner rushes.
    - **Benchmarking strategy:** How will you plan to benchmark your proposed solution before deploying to production?

6. **Business Impact & A/B Testing Design (15%) [***May be you can compare against the how base line strategy would have worked against your proposed solution***]**
    - **Metric Translation:** How logically you connect your model's offline performance (like Precision or NDCG) to projected business outcomes (like AOV lift and Cart-to-Order ratio).
    - **Experimentation:** The rigor of your proposed A/B testing framework to evaluate the model live, including guardrail metrics to ensure you aren't harming the user experience.

Submission Guidelines

To be updated shortly !!

Submission Steps

To be updated shortly !!